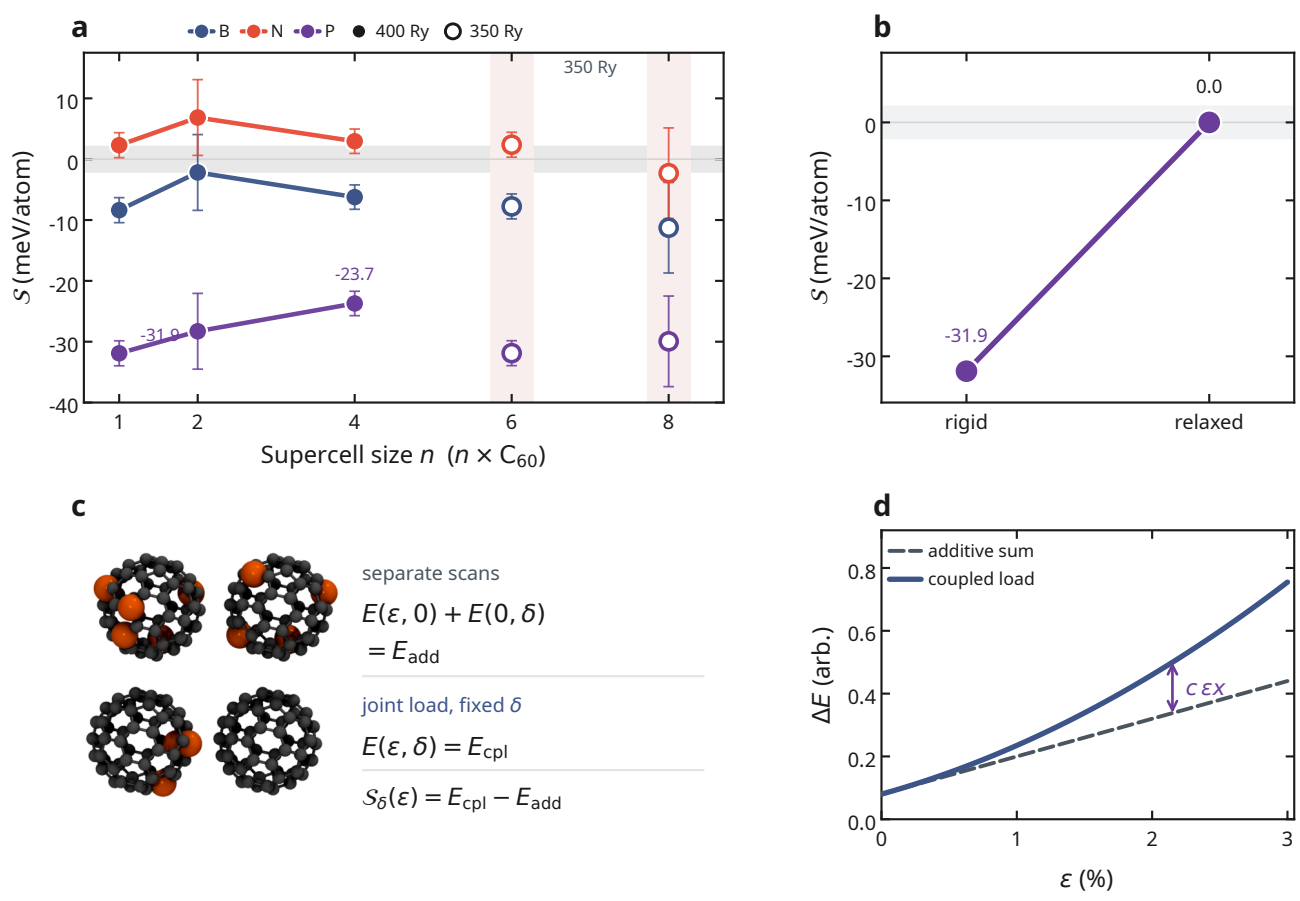

a
B
N
P
400 Ry
350 Ry
350 Ry
10
0
-10
-20
-30
-40
-23.7
-31.9
1
2
4
6
8
S (meV/atom)
Supercell size n (n × C60)
b
0.0
0
-10
-20
-30
-31.9
rigid
relaxed
S (meV/atom)
c
separate scans
E(ε, 0) + E(0, δ)
= E_add
joint load, fixed δ
E(ε, δ) = E_cpl
S_δ(ε) = E_cpl − E_add
d
additive sum
coupled load
0.8
0.6
0.4
0.2
0.0
0
1
2
3
c εx
ΔE (arb.)
ε (%)